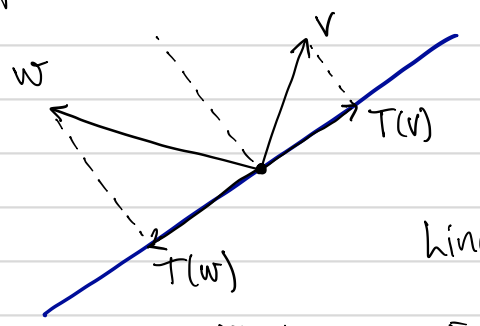
Lecture 9 - Linear Transformations

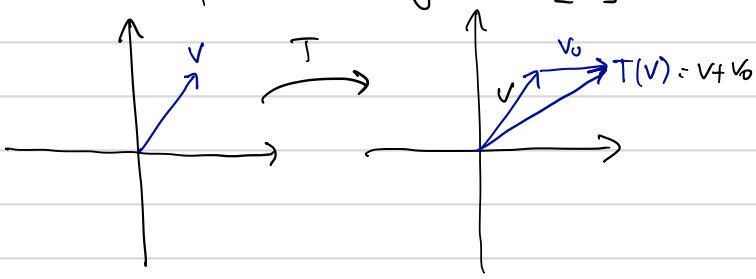
$$T(u+v) = T(v) + T(w)$$
$$T(cu) = c \cdot T(u)$$

Example: $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^2$



Linear!

(Non-)example: Shift by $v_0 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$



$$T\left(2\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

$$\neq$$

$$2 \cdot T\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = 2 \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$$

Not linear!

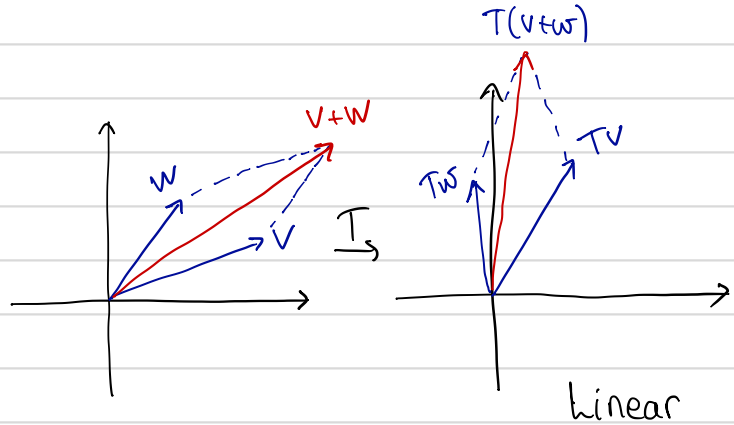
Fact: Linear transformations send zero to zero.

(Non-)Example: $T: \mathbb{R}^3 \longrightarrow \mathbb{R}^1$

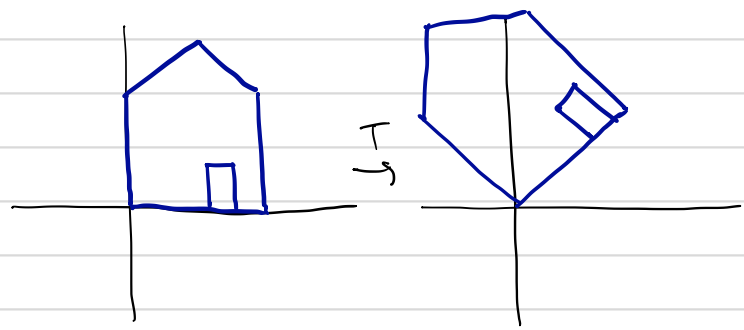
$$T(v) = \|v\|$$

length of v.

Example: Rotation by 45°



Linear

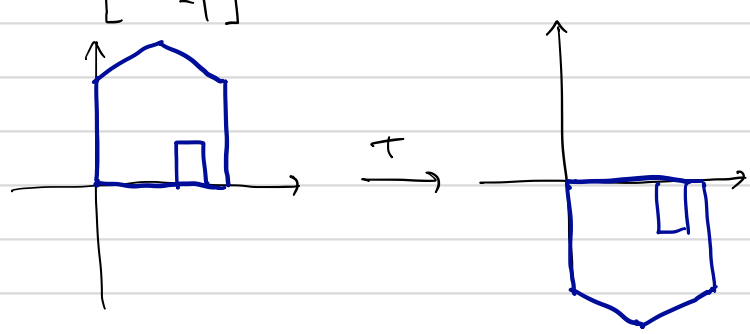


Example: Matrix A

$$T(v) = A \cdot v$$

$A(v+w) = Av + Aw$ ✓ $\quad A(cv) = c \cdot Av$ ✓

$$A = \begin{bmatrix} 1 & \\ & -1 \end{bmatrix}$$



<u>Want</u>: $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^2$

<u>Example</u>:

$$\underbrace{T(v)}_{\text{output in } \mathbb{R}^2} = \overbrace{A}^{2 \times 3}\underbrace{v}_{\text{input in } \mathbb{R}^3}$$

rows columns

Information needed to know $T(v)$ for all inputs v:

Know $T(v), T(w) \rightsquigarrow$ Know $T(2v+3w)$!

$$v = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix} = 2 \cdot \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{e_1} + 5 \cdot \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{e_2} - 1 \cdot \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{e_3}$$

$$T(v) = 2 \cdot T(e_1) + 5\, T(e_2) - T(e_3)$$